

SimpleSwitch Widget Documentation

Overview

`SimpleSwitch` is a **custom toggle switch widget** built using Flutter. It behaves like a lightweight alternative to Flutter's built-in `Switch` widget, while allowing full control over **design, animation, and colors**.

This widget is: - Reusable - Fully controlled by parent state - Animated (alignment + background) - Simple and clean UI

When to Use

Use `SimpleSwitch` when: - You want a **custom-designed switch** - You need **brand colors** instead of Material defaults - You want **smooth animations** with minimal code - You are following a **controlled widget pattern**

Widget Features

- `AnimatedContainer` for smooth background color transition
 - `AnimatedAlign` for knob movement animation
 - `GestureDetector` for tap handling
 - Stateless behavior from parent (controlled via `value`)
-

Constructor

```
const SimpleSwitch({  
  Key? key,  
  required this.value,  
  required this.onChanged,  
});
```

Parameters

Parameter	Type	Description
<code>value</code>	<code>bool</code>	Current ON/OFF state of the switch
<code>onChanged</code>	<code>ValueChanged<bool></code>	Callback triggered when switch is tapped

How It Works (Logic Explained)

1. Parent widget passes a `bool value`
2. On tap, the value is toggled:

```
widget.onChangeed(!widget.value);
```

3. UI updates based on `value`
 4. Animations handle smooth transitions
-

Usage Example

1. Parent Widget State

```
bool isEnabled = false;
```

2. Using `SimpleSwitch`

```
SimpleSwitch(  
  value: isEnabled,  
  onChanged: (newValue) {  
    setState(() {  
      isEnabled = newValue;  
    });  
  },  
)
```

Visual Behavior

State	Background Color	Knob Position
OFF	Grey (<code>cBEBEBE</code>)	Left
ON	White	Right

Full Source Code

```
import 'package:flutter/material.dart';
import 'package:nanduba/export.dart';

class SimpleSwitch extends StatefulWidget {
  final bool value;
  final ValueChanged<bool> onChanged;

  const SimpleSwitch({
    super.key,
    required this.value,
    required this.onChanged,
  });

  @override
  State<SimpleSwitch> createState() => _SimpleSwitchState();
}

class _SimpleSwitchState extends State<SimpleSwitch> {
  @override
  Widget build(BuildContext context) {
    return GestureDetector(
      onTap: () => widget.onChanged(!widget.value),
      child: AnimatedContainer(
        duration: const Duration(milliseconds: 200),
        width: 45,
        height: 18,
        padding: const EdgeInsets.all(1),
        decoration: BoxDecoration(
          border: Border.all(color: AppColors.primary),
          borderRadius: BorderRadius.circular(14),
          color: widget.value ? AppColors.white : AppColors.cBEBEBE,
        ),
        child: AnimatedAlign(
          duration: const Duration(milliseconds: 200),
          alignment: widget.value
            ? Alignment.centerRight
            : Alignment.centerLeft,
          child: Container(
            width: 20,
            height: 20,
            decoration: const BoxDecoration(
              shape: BoxShape.circle,
              color: AppColors.primary,
            ),
          ),
        ),
      ),
    );
  }
}
```

```
        ),  
      ),  
    ),  
  );  
}  
}
```

Customization Ideas

- Add `disabled` state
- Add haptic feedback
- Support RTL layout
- Make size configurable
- Add accessibility semantics

Summary

`SimpleSwitch` is a clean, animated, and reusable custom switch widget that follows best Flutter practices and is easy to extend.

You can copy, edit, and reuse this documentation anytime.